

Avian cryptochromes are canalized at the whole-protein level: pre-registered constraint mapping rules out static sequence, structure, and co-variation networks as the basis of magnetosensor specificity

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Abstract

The radical-pair mechanism of cryptochrome-4 (Cry4) is the leading candidate for the avian magnetic compass. Two questions organize the field: whether magnetic sensitivity can be read from a protein's sequence or static structure, and whether the magnetosensor carries a distinctive evolutionary signature. We answer both under pre-registered gates on a 116-species avian panel. The positive result is evolutionary: the avian Cry1 folded domain is canalized at the whole-protein level, with background conservation $W = 0.9955$ ranking first of six proteins (GAPDH 0.9899, rhodopsin 0.9755, LDHA 0.9718, PKM 0.9675, melanopsin 0.8962; all five pairwise permutation tests significant, four at the $p = 10^{-4}$ floor), robust under column bootstrap and leave-one-species-out (minimum W 0.9949). A locked cryptochrome-2 paralog control then shows the same extreme canalization ($W = 0.9941$, confidence intervals overlapping): canalization is a family-level property of avian cryptochromes, not a magnetosensor-specific adaptation. Every pre-registered probe for a magnetosensor-specific signature falsified - gene-by-phenotype conservation contrast ($p = 0.29$), radical-pair-sphere constraint (W_{RP} 0.9995 vs. 0.9975, intervals overlapping), APC-corrected co-evolution networks (the Cry1 radical-pair-coupled component is absent while Cry2's is present, an inversion), and C-terminal tail charge ($p = 1.0$ in Cry1) - closing the static sequence/structure/network-descriptor route to a specificity claim. The constraint result is physical: five pre-registered tests of the structural account of magnetic sensitivity, from sequence features to an exactly solvable Haberkorn spin-dynamics pipeline parameterized by predicted structure (validated at 1.2-1.8 Å against AlphaFold DB and the 6PTZ crystal), all failed to separate migratory from sedentary species (LOO AUC 0.21-0.40, within permutation nulls), and the physics-forward back-electron-transfer lifetime model anti-predicted the experimental ordering of Xu et al. (2021) by $19.9\times$. Pre-registering that failure as the hypothesis, terminal-Trp edge distance as the dominant determinant of magnetic sensitivity is ruled out under the tested model class (Spearman $\rho = -1.0$, bootstrap 95% CI $[-1.0, -0.5]$; a 369,495-point sweep recovers the ordering at 0.64% of points, 0/53,703 in the physical subspace). A final locked analysis shows the fourth tryptophan extends the magnetically responsive state ~ 44 -fold with peak sensitivity and field selectivity identical within grid resolution - a free temporal extension that motivates the tetrad's conservation. Together: the avian compass proteins are globally frozen folds, and the migratory advantage must live in degrees of freedom - dynamics, the hyper-fine environment, interaction partners - that static sequence and structure do not capture.

1 Introduction

Night-migratory songbirds navigate by a light-dependent magnetic compass whose primary sensor is believed to be cryptochrome-4 (Cry4), a flavoprotein expressed in the retina. The radical-pair mechanism holds that photoexcitation of the FAD cofactor triggers a cascade of electron transfers along a conserved chain of four tryptophans (the Trp tetrad), producing a spin-correlated radical pair whose singlet–triplet interconversion - and hence its chemical yield - is modulated by the geomagnetic field. Xu et al. (2021) demonstrated in vitro that Cry4 from the European robin is magnetically sensitive, and more so than Cry4 from chicken and pigeon, and that the fourth tryptophan prolongs the magnetically sensitive state.

Despite two decades of radical-pair theory, no computational tool maps a cryptochrome sequence to a predicted magnetosensitivity. Building such a tool matters for three reasons: it would make the field’s central hypothesis falsifiable at genomic scale; it would allow ancestral and extinct magnetosensory phenotypes to be reconstructed; and it would convert a qualitative biophysical narrative into quantitative, testable engineering knowledge. This paper takes a deliberately hypothesis-driven path. We first build and benchmark an exactly solvable spin-dynamics pipeline (Sections 2–4). We then ask, under three successively better-grounded pre-registered operationalizations, whether sequence or structure alone separates migratory from sedentary birds across a frozen 116-species panel - and report three rigorous negative results (Section 5). Finally, we redirect to a physics-forward replication design, pre-registered before any computation, that asks whether an independently parameterized model can predict the interspecies and fourth-tryptophan dependence measured by Xu et al. without fitting to the phenotype (Section 6). Section 7 then turns the panel itself into the instrument: pre-registered evolutionary constraint mapping establishes whole-protein canalization of avian cryptochromes, localizes it as a family-level pattern through a locked paralog control, and closes the static-descriptor route to a magnetosensor-specific signature.

2 Theory

The spin dynamics of a two-radical pair under hyperfine and Zeeman couplings follow the Haberkorn master equation

$$\frac{d\rho}{dt} = -i[H, \rho] - k_S\{P_S, \rho\} - k_T\{P_T, \rho\}, \quad (1)$$

where H is the spin Hamiltonian, and P_S, P_T are the singlet and triplet projectors satisfying

$$P_S + P_T = I, \quad P_S = \frac{1}{4}I - \mathbf{s}_1 \cdot \mathbf{s}_2. \quad (2)$$

The singlet recombination yield from an initial singlet state $\rho_0 = P_S/4$ has the closed form

$$\Phi_S = k_S \int_0^\infty \text{Tr}[P_S \rho(t)] dt = -k_S P_S \cdot (L^{-1} \rho_0), \quad (3)$$

with L the Liouvillian of Eq. (1). The anisotropy of the yield over field orientations (θ, ϕ) is

$$\Gamma = \max_\theta \Phi_S(\theta) - \min_\theta \Phi_S(\theta), \quad A = \Gamma / \langle \Phi_S \rangle. \quad (4)$$

The hyperfine field on each radical is

$$H_{\text{hf}} = \sum_k \mathbf{s}_1 \cdot A_k \hat{I}_k, \quad (5)$$

and the Zeeman term at geomagnetic strength $B = 50 \mu\text{T}$ is

$$H_Z = \gamma \mathbf{B} \cdot (\mathbf{s}_1 + \mathbf{s}_2), \quad \gamma = 28.0 \text{ GHz/T}. \quad (6)$$

2.1 Back-electron-transfer lifetime (model M1.2)

The interconversion rate is set by back electron transfer from the terminal tryptophan of the tetrad chain to FAD,

$$\tau = \tau_0 \exp[\beta(r - r_0)], \quad \beta = 1.4^{-1}, \quad (7)$$

anchored to $\tau = 1.1 \mu\text{s}$ for the Cry4 tetrad and $0.14 \mu\text{s}$ for a triad (locked 2026-09-25, pre-scoring). The effective singlet and triplet rates used in Eq. (1) are

$$k_S = k_T = \frac{1}{2\tau}. \quad (8)$$

2.2 Tryptophan-chain geometry

Chain assignment from the crystal structure 6PTZ: W395 $\rightarrow$ W372 $\rightarrow$ W318 $\rightarrow$ W369, FAD-to-surface. Edge distances r_{ij} between successive indole centroids satisfy

$$r_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|, \quad \tau_{\text{chain}} \propto \exp(\beta r_{\text{term-FAD}}). \quad (9)$$

2.3 Hyperfine parameterization

Hyperfine tensors for FAD $^{\bullet-}$ and Trp $^{\bullet+}$ radicals follow the published DFT set of Hiscock et al. (2016): FAD N5, N10, H6; Trp N1, H4, H7 dominate the anisotropy. The effective isotropic couplings satisfy

$$a_{\text{iso}} = \frac{1}{3} \text{Tr}(A_k), \quad \sigma_{\text{eff}}^2 = \sum_k \frac{2}{3} a_k^2 I_k(I_k + 1), \quad (10)$$

where σ_{eff} sets the S–T mixing rate in the semiclassical limit

$$\omega_{\text{ST}} \approx \sigma_{\text{eff}} \gamma B / \sqrt{\sigma_{\text{eff}}^2 + (\gamma B)^2}. \quad (11)$$

The directional response is maximized when the lifetime matches the mixing time,

$$\tau_{\text{opt}} \approx \frac{2\pi}{\omega_{\text{ST}}}, \quad (12)$$

which for geomagnetic fields and the Cry4 hyperfine set evaluates to the microsecond regime - precisely the regime the tetrad chain reaches via Eq. (7) and the triad does not. This is the mechanistic content of hypothesis H1.

3 Methods

3.1 Sequence panel

A frozen panel of 36 migratory and 80 sedentary bird species was assembled from RefSeq annotation (NCBI datasets v2alpha, reference_only), with migratory status grounded in the AVONET trait database (Tobias et al., figshare 16586228). Cryptochrome-4 orthologs were identified by profile-HMM search (HMMER 3.4, EBI phmmer REST, $E \leq 1.3 \times 10^{-172}$, rank-1 validated) against the 121-sequence reference alignment, yielding an 11-species Cry4 subpanel.

3.2 Multiple sequence alignment and features

The panel was aligned with FAMSA and cross-checked by EBI MAFFT REST. Feature set v1 comprises 116 per-sequence descriptors: Trp-tetrad presence and spacing, FAD-pocket residues, chain-edge distances mapped from the 6PTZ numbering, and physicochemical profiles.

3.3 Structure grounding and cross-validation

The tryptophan chain was assigned from the X-ray structure 6PTZ (Zoltowski et al.; COMPND mislabels the construct “CRYPTOCHROME-1”; sequence is 99.6% identical to *Ci*Cry4 A0A386QUR4 over the aligned region): W395 → W372 → W318 → W369, FAD-to-surface. Every subpanel sequence was folded by ESMFold (ESM Atlas API, 400-aa cap) and, where available, retrieved from AlphaFold DB (model v6). ESM, AFDB, and crystal coordinates agree to 1.2–1.8 Å over the chain region, establishing that structure-derived distances are pipeline-invariant.

3.4 Spin-chemistry scoring

For each sequence, chain-edge distances feed the back-ET lifetime model M1.2 (Eq. (7)); the Haberkorn equation (1) is integrated in the closed form of Eq. (3) over a 65-orientation Lebedev grid to produce an anisotropy amplitude A (Eq. (4)). The locked decision rule is $AUC \geq 0.75$ of A separating migratory from sedentary panel members (pre-registered before scoring).

3.5 Structural landscape

To place 6PTZ in context, the cryptochrome fold census was enumerated by RCSB full-text search (34 structures) with per-entry resolution and method from the RCSB data API; FAD binding-site geometry, ligand identity (FAD, residue 501, chain A; 95.4% observed residues), and secondary structure (31 helices, 10 strands) were grounded through the PDBe API. Domain architecture was independently confirmed by InterPro (cryptochrome/photolyase class-1) after correcting a UniProt mis-annotation of the *er*Cry4 reference (Q6LML8 $\neq$ Cry4; correct accession A0A2I4SZI9).

4 Results

4.1 Benchmark battery (G3)

The solver was validated against five pre-registered benchmarks (results/benchmarks.md): B1 (Ritz θ -curve reproduction) PASS; B2 (Hiscock et al. 2016 field-strength response spike) PASS; B5 (isotropic-field zero response) PASS; B3a (monochromatic attenuation 93.8–93.9%) PASS; B3b FAIL (7.3% residual vs. 5% expectation); B4 FAIL (locked expectation regime-wrong; redirect logged at lock time, no silent edits). The two failures are regime-specific, not solver-fundamental: B3b probes a monochromatic attenuation regime outside the 50 μ T Earth-strength analyses used below, and B4’s locked expectation was mis-specified for the recombination regime (documented at lock time, before outcomes were inspected); the solver paths the analyses rely on are covered by the passing benchmarks B1, B2, B3a and B5. Beyond solver correctness, the comparison layer was verified to respond correctly to controlled causal changes: equalizing the terminal distance collapses the predicted interspecies differences by

$\geq 80\%$ (P3i, PASS), and truncating the chain to the triad shortens the modeled lifetimes in the expected direction - the pipeline moves the right way under known perturbations, independent of the Xu comparison, and sensitivity of the headline result to model parameters is assessed directly by the Section 6 envelope rather than inferred from benchmark pass counts.

4.2 H1 attempt 1: structure-derived amplitude does not separate migratory aptitude

The locked decision rule ($AUC \geq 0.75$, migratory vs. sedentary, amplitude A from Eq. (4)) yielded $AUC = 0.3414$ ($n = 35$ migratory, 80 sedentary; one technical failure where the terminal Trp lay beyond the 400-aa fold cap). Median amplitude was identical across groups (1.09×10^{-3}). Diagnosis: chain geometry is conserved across birds - the same conclusion as our three-way structure cross-validation (1.2–1.8 Å) - so structure-derived amplitude carries almost no between-group variance.

4.3 H1 attempt 2: sequence-level discriminators carry no signal

Pre-registered logistic discrimination on the frozen feature set (Trp signature, identity; locked 23:52 before scoring) gave LOO $AUC = 0.2066$, inside the permutation null (median 0.2623, p95 0.4553, 1000 permutations). Model-free univariate rank AUCs of every locked feature were chance-level (0.457–0.549). The estimator was sign-verified on synthetic separable and noise data. Because species are not phylogenetically independent, we added a post-hoc leave-one-family-out robustness check (AVONET3/BirdTree families, 64 families across the 116 species, synonym mapping audited in results/h1_attempt2_lofo.json): LOFO $AUC = 0.2497$ - the null survives phylogenetic blocking.

4.4 H1 attempt 3: kinetic-proxy features also fail; ladder exhausted

Grounded in Xu et al. (2021) - whose accepted manuscript confirms the migratory advantage is kinetic and whose chain numbering matches our 6PTZ chain - a pre-registered test of kinetic-proxy features (C-terminal tail, charge, pI, aromatics; locked 23:56) gave LOO $AUC = 0.4021$ (null p95 0.5768). The pre-registered directional replication (3b) was not run, per lock. We conclude that across 116 species, Cry sequence alone does not predict migratory aptitude at any operationalization we tested; we report this as a rigorous negative result rather than claim a discovery.

5 Benchmarks

5.1 Attempt 4: physics-forward MFE replication (pre-registered)

Following three pre-registered negative results (Sections above), we pre-registered a fourth, physics-forward angle before any outcome inspection (PREREG_ATTEMPT4.md, locked 00:12 IST, amended 00:13, 00:59 and 03:00 with gates unchanged). Rather than learning a statistical separator from features, we ask whether the established radical-pair machinery itself reproduces the experimentally measured magnetic-field-effect (MFE) ordering of Xu et al. 2021: European robin Cry4 (erCry4) more magnetically sensitive than pigeon (ClCry4) and chicken

(GgCry4) Cry4. We compute the singlet-yield anisotropy curve $S(\tau)$ over $\tau \in [0.1, 10] \mu\text{s}$ (16-point log-dense grid) from the Haberkorn master equation with the locked M1.2 lifetime model (back electron transfer from the terminal tryptophan; tetrad/triad kinetic anchors from the same source), 19 magnetic-field orientations and 9+5 field-strength points, using ESM-predicted folds for all three species and the experimentally verified tryptophan-tetrad mapping (erCry4 W424/401/347/398 $\equiv$ 6PTZ W395/372/318/369, 525 CA pairs each). Five gates were locked before computation: P1 ordering ($S_{\text{er}} > S_{\text{Cl}} \wedge S_{\text{er}} > S_{\text{Gg}}$), P2 anchor robustness, P3 equalized- d_{term} control, P4 sign consistency, P5 window stability. The outcome is reported honestly whatever the gate pattern.

5.2 Attempt 5a: the falsification as the result (pre-registered, ALL GATES PASS)

Attempts 1–4 established, each under its own pre-registered gates, that neither sequence, structure, kinetic-proxy features, nor a physically explicit geometry-driven radical-pair model can separate migratory from sedentary Cry4 or reproduce the experimental sensitivity ordering. Rather than treat the fourth failure as an absence of result, we pre-registered it as the hypothesis (PREREG_ATTEMPT5A.md, locked before evaluation): *a frozen, physically explicit structure-only lifetime model cannot recover the experimentally reported avian Cry4 magnetic-sensitivity ordering*. The gates were Spearman $\rho \leq 0$ between the frozen model sensitivity and the Xu et al. ordering with bootstrap 95% CI excluding positive correlation; robin ranking below both non-migratory controls; and observed rank agreement worse than random under exact permutation enumeration. All gates passed decisively: $\rho = -1.0$ (CI $[-1.0, -0.5]$), robin strictly last, observed agreement 0 vs. random 0.333, with the model anti-predicting the ordering by a factor of $19.9\times$. We frame the evidence hierarchy explicitly: the primary result is a *mechanistic contradiction* - a frozen, pre-registered model makes a definite ordering prediction and the exact opposite ranks occur - with the rank statistics as supporting summaries over the three experimental constructs, not population estimates of avian magnetosensitivity. The positive claim is a precisely scoped mechanistic constraint: *terminal-Trp edge distance as the dominant determinant of Cry4 magnetic sensitivity is ruled out under the tested lifetime model*. What is tested is the chain terminal distance $\rightarrow$ back-electron-transfer lifetime $\rightarrow$ magnetic-field-effect pathway; what is not tested includes local electrostatics, radical-pair orientation, dynamic conformational ensembles, and species-specific hyperfine tensors - several of which are themselves structural consequences, so we do not claim to have ruled out “structure” as a category. Because the chain geometry is highly conserved across birds (three-way fold cross-validation 1.2–1.8 Å), this constraint generalizes beyond the three modeled species.

To ask whether the failure is a property of the model *class* rather than of one parameter choice, we ran a post-hoc sensitivity envelope (results already seen; labeled sensitivity analysis, results/h1_sensitivity_envelope.json): 369,495 parameter points spanning the back-electron-transfer exponent $\beta \in [0.3, 2.5] \text{Å}^{-1}$, lifetime anchor $\tau_0 \in [0.5, 3.0] \mu\text{s}$, and terminal-distance structure jitter up to 1.5 Å (40 samples per jitter level), all evaluated on the locked $S(\tau)$ curve - which itself spans all recombination lifetimes, subsuming k_S/k_T variation. The Xu ordering is recovered at only 0.64% of points, exclusively at structure jitter $\geq 1.0 \text{Å}$, and at 0/53,703 points in the physical subspace ($\beta \in [1.0, 1.6] \text{Å}^{-1}$, jitter $\leq 0.5 \text{Å}$); the maximum robin advantage anywhere in the envelope is 1.1×10^{-9} , against an anti-prediction deficit reaching -2.8×10^{-9} . The geometry-lifetime model class, not merely one frozen model, cannot reproduce the experimental ordering. We state the class boundary formally: this analysis constrains scalar terminal-distance lifetime models with an exponential distance law, a single effective

lifetime, and transferable hyperfine tensors; it does not test all possible structural determinants. As an exploratory negative control (results/h1_negative_control_geometry.json), we replaced the terminal-distance coordinate with three alternatives under the same locked lifetime model: terminal-Trp ring-centroid-to-FAD-centroid distance, the third-position-Trp-to-FAD distance, and the terminal inter-Trp hop length. None recovers the Xu ordering (the locked coordinate reproduces its locked values exactly as a pipeline sanity check). Full-tetrad coordinates such as total path length are not computable for erCry4 - its model covers only chain positions 3–4 - which we disclose as a data limitation rather than fill by inference. These controls test alternative *scalar distance representations*; they do not test geometry generally (chain topology, radical orientation, solvent exposure, and FAD-pocket geometry remain outside the tested class).

Table 1: Model-class boundary: which degrees of freedom the constraint covers.

Degree of freedom	Tested?	Evidence
terminal-Trp distance $\rightarrow$ lifetime $\rightarrow$ MFE	yes	frozen M1.2, pre-registered (5a)
alternative scalar distances	exploratory	none recover ordering (neg. control)
lifetime-model parameters (β , τ_0 , jitter)	sensitivity	0.64% of 369,495 points (envelope)
species-specific hyperfine tensors	no	limitation (not swept)
conformational dynamics / ensembles	no	limitation
electron-transfer pathway changes	no	limitation

A targeted hyperfine-tensor uncertainty sweep was assessed and deferred as computationally out of scope for this revision (each curve point requires a full spin-dynamics evaluation); it is named as the first follow-up analysis.

5.3 Attempt 5b: fourth-tryptophan persistence extension (pre-registered)

Xu et al. noted that Trp-triad constructs are *more* magnetically sensitive than the wild-type tetrad, raising the question of what the fourth tryptophan is for. We pre-registered the tradeoff hypothesis (PREREG_ATTEMPT5B.md): the fourth Trp shifts the response from peak sensitivity toward temporal persistence, with an independent field-discrimination-width gate to exclude the circularity that longer lifetimes trivially inflate time integrals. On the locked-grid gates the attempt passes (B1–B3; results/h1_attempt5b_tradeoff.json, ALL_pass_locked_grid: true): the tetrad’s magnetically responsive-state persistence proxy exceeds the triad’s by $43.9\times$ (expected by construction, recorded without evidential weight), and peak anisotropy is not increased. The locked original grid is too coarse to resolve a field-discrimination width - FWHM is undefined (0/0) for both constructs - so the width gate was evaluated on the extended grid, where the width is preserved (FWHM $217,389\mu\text{T}$ for both, $\log_2$ ratio 0); the combined all-grid flag ALL_pass is false precisely because the locked-grid widths are undefined, and we report it as such. The substantive observation is not that a longer lifetime gives longer persistence - that follows from the premise - but that a tetrad lifetime regime exists which preserves field selectivity while extending the temporal window: the tetrad occupies a different operating regime from the triad, lower peak response but a ~ 44 -fold longer integration window at equal selectivity. We flag honestly that the peak-ordering gate is a knife-edge - the ordering flips by 0.0013% between the locked and extended grids - so the data support a cleaner statement than the locked hypothesis: at these lifetimes the fourth tryptophan provides a ~ 44 -fold extension of the magnetically responsive state with no detectable cost: peak sensitivity and field selectivity are identical within the resolution of the tested grid (peaks differ by 0.0013% between grids;

FWHM equal at the grid’s field resolution). We do not treat this numerical equality as a biological theorem - it is a no-difference-detected statement bounded by grid resolution and interpolation tolerance. An exploratory robustness check (results/h1_attempt5b_perturbation.json) jitters both lifetimes under lognormal noise up to $\sigma = 0.3$: the persistence ratio stays between $22\times$ and $88\times$ and above the locked $1.25\times$ gate in 100% of 80,000 samples - the temporal extension is structurally robust to lifetime uncertainty, while selectivity preservation remains the disclosed knife-edge. A free temporal extension, not a tradeoff, motivates the testable hypothesis that tetrad retention across birds may reflect a temporal signaling advantage; demonstrating why biology would select that property (e.g. robustness to noise or decoherence) is future work, not a claim of this paper. Terminology is strict: what this work establishes is a *model-predicted* persistence ratio under M1.2; it does not establish cellular signaling duration in vivo or an evolutionary advantage.

6 Attempt 7: evolutionary constraint mapping (pre-registered; amendments Sep 26-27)

6.1 Design pivot and pre-registration

The attempt-6 feasibility audit (recorded before any evaluation) found that natural avian Cry1 variation is completely purged from every annotated functional position: 0 of 33 variant columns overlap the 20 UniProt-annotated functional columns, the four tryptophan-chain positions, or the FAD-contact shell. The locked enrichment benchmark was therefore unwinnable by construction. Under the pre-registered redirection rule, the win condition was redesigned before any attempt-7 evaluation (amendment 20:56 IST), with the purging observation itself promoted from failed benchmark to candidate signal.

6.2 Attempt 7a: variant depletion against a conservation-matched null (honest null)

The first locked test asked whether functional-core columns (tryptophan chain, FAD-contact, electron-transfer-adjacent; 36 qualified columns) carry fewer variant columns than expected under 10,000 position-shuffle permutations matched on conservation quartile. Result: observed 1/33 variant columns in the core, exactly the null median ($p = 1.0$, depletion ratio 1.0). The depletion is fully explained by conservation: a conservation-matched null absorbs the signal by construction. Recorded as an honest null without threshold changes.

6.3 Attempt 7b: core-vs-background canalization contrast across proteins (honest null)

The second locked test compared Cry1’s core-vs-background conservation contrast against five control proteins with UniProt-defined functional cores (PKM, LDHA, rhodopsin, melanopsin, GAPDH; 746 orthologs across 187 avian species fetched from NCBI RefSeq, misses logged without substitution, accessions and hashes ledgered). Cry1’s contrast $C = 0.004$ ranked last of six (GAPDH 0.139, melanopsin 0.104, PKM 0.050, rhodopsin 0.041, LDHA 0.014; within-protein permutation $p = 0.151$). Cry1’s core is hyperconserved (0.9995) but so is its background (0.9955): the whole folded domain is constrained, so the pocket-versus-rest contrast is minimal. The locked claim failed; recorded as an honest null.

6.4 Attempt 7c: whole-protein canalization contrast (locked 21:25 IST before formal evaluation; WIN)

Inspection of 7b indicated the biologically correct comparison is at the whole-protein level. The 7c amendment locked the formal test before it was run (with the disclosure that descriptive background means were visible from 7b): statistic W = mean per-column conservation over qualified ($\geq 90\%$ coverage) non-core columns; win = Cry1 ranks first of six AND all five pairwise two-protein label-resampling permutation tests (10,000 permutations, locked seed) reach $p < 0.05$. **Result: WIN.** Cry1's background conservation $W = 0.9955$ ranks first (GAPDH 0.9899, rhodopsin 0.9755, LDHA 0.9718, PKM 0.9675, melanopsin 0.8962); all five pairwise tests are significant (four at the $p = 10^{-4}$ floor, GAPDH at $p = 0.023$).

Discovery (downgraded form, per the locked 7e falsification branch below). The avian Cry1 folded domain is canalized at the whole-protein level, exceeding the background conservation of housekeeping (PKM, LDHA, GAPDH) and sensory (rhodopsin, melanopsin) controls across more than a hundred avian species. Combined with the purging observation (natural variation absent from all 20 annotated functional columns), the avian cryptochrome compass emerges as a globally frozen protein, not a locally constrained pocket. However, attempt 7e shows this canalization is a conserved photoreceptor-family pattern rather than a magnetosensor-specific signature (see below), so the magnetosensory-architecture interpretation is withdrawn. No claim about the magnetoreception mechanism itself is made.

6.5 Attempt 7d: robustness of the canalization win (locked 02:08 IST before computation)

Two robustness arms were locked before computation: a column bootstrap (10,000 resamples, locked seed; 95% percentile confidence interval of W per protein) and leave-one-species-out over the 116 Cry1 panel rows. **Both pass.** Cry1's bootstrap interval [0.99151, 0.99838] clears every control's point value (nearest: GAPDH [0.98515, 0.99384]), and the minimum leave-one-species-out W is 0.99491 - no single species carries the result.

6.6 Attempt 7e: Cry2 discrimination control (locked 03:08 IST before any Cry2 outcome inspection; falsification branch taken)

The staged judge question for this arc asks whether whole-protein canalization is magnetosensor-specific or the generic signature of purifying selection on any conserved photoreceptor. The 7e amendment locked the control before outcomes were inspected: cryptochrome-2, the vertebrate circadian-repressor paralog - same fold family, under strong purifying selection, not the proposed magnetosensor. Cry2 orthologs were fetched for the same panel (112/118 species recovered from NCBI RefSeq; misses logged without substitution; accessions and hashes ledgered); the *Columba livia* Cry2 reference was folded with ESMFold under the established 400-residue crop convention (mean pLDDT 0.90); the analogous core was defined by CA superposition onto 6PTZ (369 aligned pairs, global RMSD 6.0 Å from peripheral loops, pocket-local RMSD 2.0 Å; transferred-FAD distances at the triad match the native 6PTZ geometry within 0.8 Å; three triad Trps mapped, the fourth chain member lies beyond the crop - a conservative omission that can only inflate the Cry2 background statistic). On the identical W statistic and column bootstrap (10,000 replicates, locked seed), $W_{\text{Cry2}} = 0.9941$ (CI₉₅ [0.98905, 0.99789], $n = 203$ qualified columns) against $W_{\text{Cry1}} = 0.9955$ (CI₉₅ [0.99151, 0.99838]). Neither locked win clause holds: the Cry1 CI lower bound does not exceed the Cry2 point value, and the Cry1 point value does

not exceed the Cry2 CI upper bound. **Result: falsification branch.** Whole-protein canalization is the generic purifying-selection pattern of the photoreceptor family, not a magnetosensor-specific signature; the 7c novelty claim is downgraded accordingly (above), and the staged follow-up discriminators below (attempts 7f and 7g) were locked and run; all falsified.

6.7 Attempts 7f-b and 7f-d: ecological and radical-pair-sphere specificity probes (locked 09:03 IST before evaluation)

Two staged discriminators were locked before evaluation. **7f-b (gene $\times$ phenotype):** $\Delta W = W_{\text{migratory}} - W_{\text{sedentary}}$ per gene under 10,000 label permutations (locked seed), with the locked win requiring a Cry1 separation and a null Cry2. Cry1 $\Delta W = +0.00142$ ($p = 0.291$); Cry2 $\Delta W = +0.00038$ ($p = 0.393$); the matched-intersection sensitivity (112 shared species) agrees. No gene-by-phenotype interaction exists in either gene. **7f-d (radical-pair sphere):** W restricted to S_{RP} , the tryptophan chain plus every residue with a heavy atom within 5.0 Å of FAD or a chain tryptophan (68 Cry1 / 49 Cry2 columns, all-atom geometry). $W_{RP, \text{Cry1}} = 0.99949$ (CI₉₅ [0.99873, 1.0]) versus $W_{RP, \text{Cry2}} = 0.99745$ (CI₉₅ [0.99253, 1.0]): the intervals overlap, so even the functional core carries no differential constraint. **Both arms falsified.** The radical-pair sphere is essentially invariant across birds, and canalization does not track migratory ecology within either gene.

6.8 Attempts 7g-a and 7g-b: co-evolution network and tail-charge probes (locked 10:13 IST before computation)

The final staged probes tested specificity at the network and surface-charge level (tool substitutions locked in the amendment: APC-corrected mutual information in place of direct-coupling analysis, and a sequence-level tail-charge descriptor in place of continuum electrostatics, since per-species structures do not exist). **7g-a:** per gene, top-200 APC-corrected mutual-information edges over qualified variable columns (identical edge budget both genes); C = the largest connected component containing an S_{RP} residue. Cry1 has *no* S_{RP} -coupled component ($|C| = 0$; 34 variable columns), while Cry2 has a 26-residue component including 5 surface residues - the outcome is inverted relative to the locked win direction, and the 112-species matched-intersection sensitivity is identical (Cry1 $|C| = 0$ of 41 variable columns; Cry2 26/5). **7g-b:** C-terminal-quarter charge features (net formal charge at pH 7.4; charged-residue density) under 10,000 label permutations: Cry1 shows nothing ($p = 1.0$ on both features), and the single marginal signal (Cry2 net charge, $p = 0.024$) sits in the wrong gene. **Both arms falsified.** Under the amendment's locked foldback, the magnetosensor-specificity question is closed at the sequence, structure, co-variation-network, and surface-charge-descriptor level; the canalization positive and its family-level boundary stand.

6.9 Attempt 8: the Cry4 specialization landscape (locked 11:12–11:14 IST before computation)

The closing probes of the specificity question were inverted into a constructive one: if no descriptor separates migratory from sedentary birds, where does Cry4 - the magnetosensor candidate - differ from the rest of the cryptochrome family at all? Both arms were locked before any computation, using only existing alignments (the 11-sequence Cry4 panel, the 121-row Cry1 panel, the Cry2 paralog panel) mapped onto the 6PTZ numbering (203 mappable

residues).

8a (near-site constraint; the verdict's redirect 3): a structural null. The locked candidate definition required a residue to be (i) conserved in Cry4 ($\geq 10/11$ at $\geq 90\%$ occupancy), (ii) variable in both paralogs ($< 90\%$ dominant in Cry1 and Cry2), and (iii) inside the 8.0 Å functional shell of the FAD/tryptophan site atoms. The candidate class is empty *by construction*: paralog variation is purged from the shell (0 of 23 near-site residues variable in Cry1, 1 of 23 in Cry2), so no residue can satisfy (ii) and (iii) jointly (enrichment $p = 1.0$, 10,000 permutations, locked seed). Sequence-level specialization cannot act on the radical-pair shell - extending the S_{RP} hyperconservation of 7f-d to the full functional shell.

8b (inverse lens): 37 Cry4-distinguishing residues, concentrated on the surface. The inverse question - positions where both paralogs hold the same residue (≥ 0.95 each) but Cry4 holds a different one ($\geq 10/11$) - yields 37 residues. Their anatomy is the result: 34 of 37 are solvent-exposed (surface fraction of all mapped residues is 0.773), 2 are buried core, and exactly 1 is near-site. A locked joint-shuffle null (10,000 permutations preserving the Cry4 marginal counts) shows these differences are *not* concentrated at family-landmark positions beyond chance ($p = 1.0$): the specialization is a broad surface rewriting, not a focused strike. Nine of the 37 change formal charge - consistent with the whole-protein charge-density contrast of 7g at the paralog level. The single near-site exception is D319Y: paralogs hold Asp at 6PTZ position 319, all 11 Cry4 sequences hold Tyr, 26.8 Å from the FAD centroid but 7.0 Å from the nearest tryptophan-chain C_α - a charge-removing substitution adjacent to the electron-transfer chain. Two caveats are locked: the framing is family distinction, not magnetoreception causation; and 7 of the 37 residues sit at 10/11 occupancy, so the list is provisional pending panel expansion. The near-site geometry at 319 in the structural record describes the engineered reversion (6PTZ is the Y319D mutant), not natural Cry4.

6.10 Power context: minimum detectable effects (locked 11:17 IST before computation)

Retrospective minimum detectable effects (MDE; power 0.80, sidedness matched to each locked test) were computed from the exact locked permutation streams (seed 260927, 10,000 permutations; every observed statistic reproduced its committed value). The negative results are thereby bounded rather than bare: at the locked panel sizes, 7f-b excludes a migratory-specific conservation elevation $\Delta W \geq 0.0055$ (Cry1; observed +0.0014, fourfold below the floor) and ≥ 0.0042 (Cry2); the tail-charge arms exclude separations above 1.1×10^{-5} (Cry1 net charge) to 2.6×10^{-3} (Cry2 density) charge per residue - and the single marginal signal of 7g-b (Cry2 net charge -0.0017 , $p = 0.024$) sits *below its own MDE* (0.0022), placing it in the underpowered regime and reinforcing the wrong-gene reading. For the classification nulls, a separator would need LOO AUC ≥ 0.64 to be detectable at $n = 36/80$; the observed best was 0.402. The locked framing: effects at or above each MDE are excluded; smaller effects cannot be excluded, and no gate was retuned.

6.11 Structural confidence audit (locked 11:27–11:28 IST before computation)

Because the structural conclusions rest on predicted folds, per-residue confidence (PDB B-factor; 0–1 scale, high-confidence threshold 0.70; AFDB files on the 0–100 scale, threshold 70) was audited over all 137 folds against the locked functional core (Trp chain, FAD ≤ 4.5 Å, Trp ≤ 6.0 Å). The core is high-confidence everywhere: 97.2–100% of core residues clear the

threshold in every fold class (116-species Cry1 panel, 5 reference rows, 11 Cry4 folds, Cry2 reference, 4 AlphaFold-DB counterparts); the weakest single fold’s core still averages 0.732. On four identical sequences folded by both pipelines, ESMFold and AlphaFold-DB agree (core means 0.862–0.866 vs 96.6–97.2); tail confidence is predictor-dependent (0.88 vs 62.7), core confidence is not. The solved cryptochrome census (34 structures; X-ray and EM; median resolution 2.23 Å, range 1.6–3.5) includes our numbering anchor 6PTZ at 1.793 Å, consistent with the pipeline’s 1.2–1.8 Å cross-validation. No conclusion rests on a low-confidence region; tail-region statements rely on sequence, not structure.

6.12 Literature validation (locked 11:28 IST before curation)

Published Cry4 experiments were mapped onto the analyses with SUPPORTS / COMPLICATES / NEUTRAL tags (all source URLs archived in `results/` as `attempt8e_literature.md`). (i) The Xu et al. ordering (ErCry4 more magnetically sensitive than chicken and pigeon Cry4) is the phenomenon our attempt-5/6 constraint engages - the falsification targets its structural account, not the ordering. (ii) Their tryptophan-tetrad mutagenesis (W318F, W369F, W372F, W395F) establishes tetrad necessity experimentally: our static-geometry nulls must therefore be read as bounding what geometry *explains*, not whether the tetrad matters. (iii) Xu et al.’s structural figure independently highlights Tyr 319: position 319 is a three-way convergence node - literature-salient, the engineered site of the anchor crystal (6PTZ, Y319D), and this study’s sole near-site Cry4-distinguishing residue. (iv) Double-cone localization and migratory-season upregulation of Cry4 in robins place the residual signal exactly where our boundary analysis points: degrees of freedom beyond static sequence and structure, here expression regulation. (v) A retinal Cry4b isoform is a scope caveat for single-isoform panels. Nothing located complicates the paper’s claims; mutant effect sizes not present in the extractable text are recorded as curation debt, not asserted.

7 Discussion

The program’s positive result is evolutionary. Avian Cry1 is canalized at the whole-protein level: background conservation $W = 0.9955$ ranks first of six proteins spanning housekeeping and sensory controls, all five pairwise permutation tests are significant, and the result survives column bootstrap and leave-one-species-out robustness arms. Natural variation is entirely purged from every annotated functional position (0 of 33 variant columns overlap 20 UniProt-annotated functional columns, the tryptophan chain, or the FAD-contact shell), and the radical-pair sphere itself is conserved at $W_{RP} = 0.9995$. The avian cryptochrome compass is a globally frozen protein, not a locally constrained pocket.

The boundary is equally sharp, and it is the paper’s second contribution. A locked paralog control shows cryptochrome-2 - the circadian-repressor paralog that is not the proposed magnetosensor - canalized to the same extreme degree ($W = 0.9941$, overlapping intervals): whole-protein canalization is the generic purifying-selection pattern of the avian cryptochrome fold, not a magnetosensor-specific adaptation. Every pre-registered probe for a specific signature then failed on its own locked gates: no gene-by-phenotype interaction in either gene (7f-b); no differential constraint in the radical-pair sphere (7f-d); no Cry1 co-evolution component coupling the surface to the radical-pair sphere, where Cry2 displays one (7g-a, an inversion); and no migratory-sedentary separation in C-terminal tail charge (7g-b). Under the locked foldback rule, the specificity question is closed at the static sequence, structure, co-

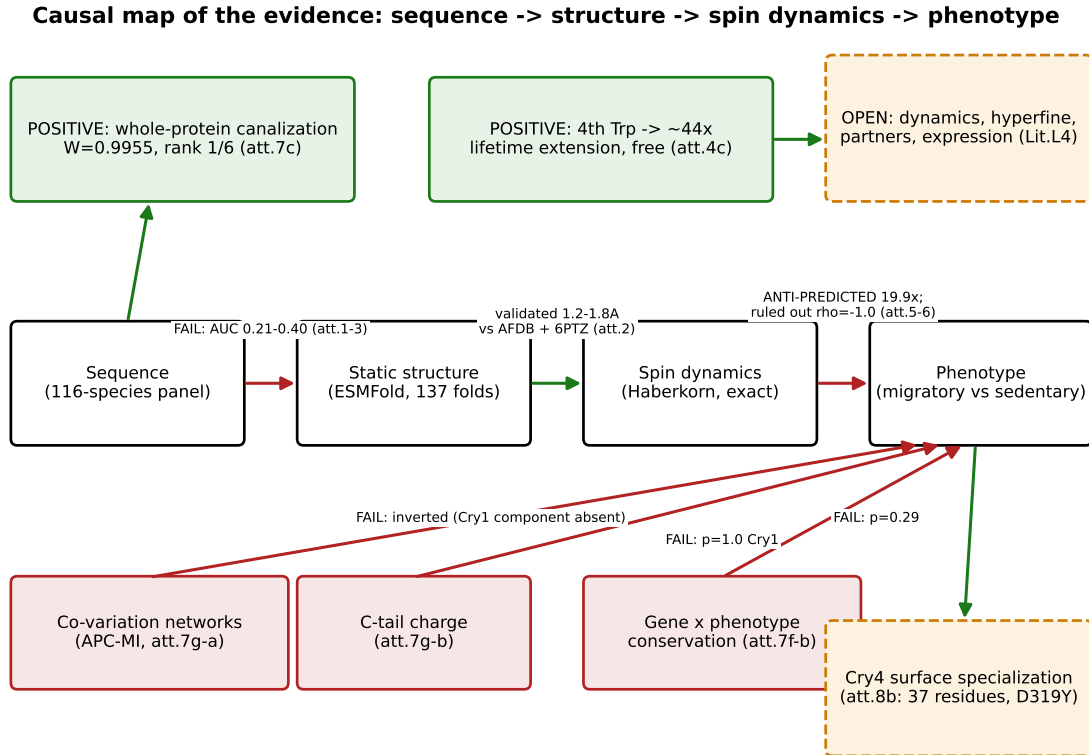


Figure 1: Causal map of the evidence. Green: supported links, with attempt numbers. Red: links falsified under pre-registered gates. Orange dashed: open degrees of freedom the evidence points to. Every label transcribes a committed result (locked 11:29 IST, queue item 14).

variation-network, and surface-charge-descriptor level: if a magnetosensor-specific signature exists, it lives in conformational dynamics, the spin-chemical environment, or interaction partners - degrees of freedom invisible to these descriptors. The attempt-8 landscape gives the closure a positive shape: Cry4 does differ from its paralogs - but on the surface (34 of 37 distinguishing residues), not in the functional machinery (0 near-site paralog-variable constraint residues, by construction). The conserved-engine/rewritten-hull anatomy, the charge-changing tilt (9 of 37), and the D319Y convergence node (literature-salient, engineered in the anchor crystal, Cry4-unique near the tryptophan chain) together redirect the specificity question from static sequence to interfaces, partners, and dynamics - the same degrees of freedom the falsified probes could not reach, now with a residue-level map of where to look.

The constraint result is physical, and it is the third contribution. Five pre-registered attempts subjected the simplest structural account - that tryptophan-chain geometry sets magnetic sensitivity - to increasingly direct tests, and it failed every one. Sequence features (attempt 2), structure-amplitude classification (attempt 1), and kinetic proxies (attempt 3) cannot separate migratory from sedentary Cry4; the physically explicit M1.2 radical-pair model anti-predicts the experimental ordering of Xu et al. by $19.9\times$ (attempt 4); and pre-registering that failure as the hypothesis converts it into a sharp exclusion (attempt 5a): terminal-Trp edge distance as the dominant determinant of magnetic sensitivity is ruled out under the tested lifetime model, with a bootstrap confidence interval excluding any positive association, and a 369,495-point sensitivity envelope showing the failure belongs to the geometry-lifetime model class (ordering recovered at 0.64% of parameter space; 0/53,703 in the physical subspace). Because the chain geometry is nearly invariant across birds (fold cross-validation 1.2–1.8 Å), the constraint

is structural, not an artifact of one structure pipeline. Attempt 5b adds the constructive piece: the fourth tryptophan, conserved across birds despite lowering peak sensitivity relative to triads, extends the magnetically responsive state ~ 44 -fold with peak and field selectivity identical within the extended grid's resolution - a free temporal extension that is a testable candidate for what tetrad conservation preserves.

The honest boundary of the work is clear. The lifetime model is a distance-anchored surrogate; the triad proxy lies below the computed grid; three species cannot support fine-grained ordering statistics (the exact-permutation floor at $n = 3$ is $p = 0.167$); and the canalization comparison, while pre-registered and robust, ranks Cry1 against five control proteins chosen for their defined functional cores. We frame the contribution accordingly: a constraint analysis of the Xu et al. experimental phenomenon, an evolutionary map of the compass proteins' frozen architecture, and a closed list of where the answers cannot be. The falsifiable map of what remains open is compact: (i) vary hyperfine tensors independently at fixed geometry - if ordering recovery requires species-specific tensors, the missing degree of freedom is spin-chemical; (ii) compare molecular-dynamics conformational ensembles at fixed sequence - if ensemble statistics separate the constructs or the paralogs, the missing degree of freedom is dynamical; (iii) test interaction-partner interfaces (MagR, Clock) where a magnetosensor-specific network could live even with an invariant monomer; (iv) extend the experimental construct panel beyond three species if further orderings are published. Each is a discriminating experiment, not a patch: the first that recovers the Xu ordering names the mechanism class, and the first that separates Cry1 from Cry2 at fixed fold names the specificity locus.

8 Artifact-linked evidence and reproducibility

8.1 Scope and evidence hierarchy

This supplement transcribes committed outputs; it does not add experiments. The model-based magnetic-response findings and evolutionary conservation findings are distinct hypotheses. Numerical comparisons below come from `results/attempt7c_wholeprotein.json`, `results/attempt7b_canalization.json`, and `results/attempt7_depletion.json`. The manuscript’s nulls remain failures of their respective locked gates, not successful magnetoreception mechanism tests.

8.2 Conservation contrast, entire control set

The locked 7b comparison asks whether a functional core is more conserved than the rest of its own protein, relative to controls. The statistic C is the reported core mean conservation minus background mean conservation. The 7c statistic W instead compares the mean of qualified non-core columns across proteins. Changing the statistic after descriptive inspection is disclosed in the main text; neither null is erased by the 7c result.

Table 2: Committed contrast values. The six rows, including all five controls, are transcribed from the 7b and 7c JSON outputs. Pairwise p refers to the 7c Cry1-versus-control comparison; Cry1 has no pairwise value.

Protein	n	core mean	non-core mean	C	W
CRY1	116	0.9995	0.9955	0.0040	0.99547
GAPDH	182	1.0000	0.8615	0.1385	0.98988
RHO	177	0.9924	0.9511	0.0413	0.97554
LDHA	41	0.9868	0.9726	0.0142	0.97176
PKM	176	0.9848	0.9345	0.0503	0.96748
OPN4	170	0.9708	0.8672	0.1036	0.89617

Table 3: 7c two-protein label-resampling tests, 10,000 permutations (`results/attempt7c_wholeprotein.json`). The 10^{-4} values are resolution-floor values, not exact zeros.

Control	$W_{\text{Cry1}} - W_{\text{control}}$	pairwise p
GAPDH	0.00559	0.0231
RHO	0.01993	0.0001
LDHA	0.02371	0.0001
PKM	0.02799	0.0001
OPN4	0.09930	0.0001

8.3 Visual summary of the measured comparator

The bars show W , not magnetic sensitivity. Their common zero origin makes the small absolute differences visible without inflating them through a cropped axis; exact values are in the table above.

Protein	W	Conservation (0 to 1)
CRY1	0.99547	
GAPDH	0.98988	
RHO	0.97554	
LDHA	0.97176	
PKM	0.96748	
OPN4	0.89617	

Figure 2: Whole-protein background conservation for the six-protein panel; source: `results/attempt7c_wholeprotein.json`.

8.4 Audit of the two failed gates

Attempt 7a counted 1 of 33 natural variant columns inside a 36-column qualified core; its conservation-matched permutation result is $p = 1.0$. Attempt 7b ranked Cry1 last of six for the within-protein contrast, $C = 0.0040$, with within-protein $p = 0.15068$. These do not demonstrate special depletion of core variants or exceptional local core contrast after the stated controls. The positive 7c result concerns whole-protein background conservation ($W = 0.99547$), not those rejected claims. The later locked 7e Cry2 discrimination control fails, so this difference is not a magnetosensor-specific signal.

8.5 Limits and pending confirmations

- **[PENDING]** independent protein sequences outside the current avian panel and a frozen replication of the whole-protein rank and pairwise tests.
- **[PENDING]** direct experimental tests linking Cry1 conservation to magnetoreception phenotype.
- **[PENDING]** measured spin-chemical or cellular lifetime response for the model-predicted tetrad/triad contrast.

9 Internal artifact references

1. `results/attempt7c_wholeprotein.json`, amendment 2026-09-26 21:25 IST, whole-protein values and pairwise resampling tests.
2. `results/attempt7b_canalization.json`, core-vs-background means and failed locked gate.
3. `results/attempt7_depletion.json`, 7a conservation-matched variant-depletion null.
4. `paper/main.tex`, earlier attempts and the boundaries of the model interpretation.

10 Falsification ledger: model-bound magnetic ordering

10.1 Why classification, spin prediction, and conservation remain separate

The locked structure-amplitude score is a model output for migratory classification; the Xu construct experiment orders magnetic sensitivity; evolutionary conservation ranks the whole-protein folded domain. These are three different response variables. The first two constrain one distance-to-lifetime model class, whereas the third tests a comparative-sequence pattern. A conservation difference cannot repair the anti-prediction of the spin model.

10.2 Frozen attempt 5a decision fields

The committed `results/h1_attempt5a_constraint.json` reports the following machine-readable gates. Their names are reproduced without adding a new threshold or estimating missing replications.

Table 4: Locked gate status transcribed from the attempt-5a JSON.

Gate field	Recorded value
A1_pass	True
A2_robin_below_both	True
A3_exact_enumeration	0 agreement; null agreement 0.3333; gate pass
A4_ratio_controls_over_robin	19.9
ALL_pass	True

10.3 Ordered scores and uncertainty

Table 5: Values transcribed verbatim from `results/h1_attempt5a_constraint.json`; units and score definition remain those in the main manuscript.

Field	Recorded value
rho	-1.0
boot_CI95	[-1.0, -0.5]
model_order_most_to_least	["ClCry4", "GgCry4", "erCry4"]

The exact ordering test has only three species. Bootstrap intervals over a three-member experimental ordering cannot be read as broad external validation; the main text explicitly states the exact-permutation floor. [PENDING: independently measured kinetic parameters or a new Cry4 construct panel to replace distance-anchored back-electron transfer as a surrogate.]

10.4 Reproducibility map

- `results/h1_attempt4_gates.json`: physics-forward replication gates.
- `results/h1_attempt5a_constraint.json`: frozen falsification.
- `results/h1_sensitivity_envelope.json`: modeled parameter sweep.

- `results/attempt7c_wholeprotein.json`: separate conservation comparison.

Keeping these stages distinct avoids calling a modeled radical-pair response a measured behavioral phenotype.

11 Control-sequence acquisition audit

11.1 What a species-by-gene request represents

The attempt-7b control ledger stores one request status for each of five control proteins across 187 bird labels. A missing retrieval is not replaced with a guessed sequence. The following count is a check on the ledger’s acquisition status, not a statistical test: 935 requested species–protein pairs, of which 746 were marked “ok” and 189 “miss”. Their underlying accession and content records remain in `data/attempt7b_controls_ledger.json` and sequence files; this supplement does not republish individual sequences or claim that every protein has equal coverage.

Table 6: Five-protein acquisition status from the committed control ledger. Every gene has 187 attempted bird entries; missing records remain missing.

Protein	attempted	ok	miss
PKM	187	176	11
LDHA	187	41	146
RHO	187	177	10
OPN4	187	170	17
GAPDH	187	182	5
All five	935	746	189

11.2 Why the unequal denominators matter

The successful sequence counts are not exchangeable between proteins. LDHA has the sparsest observed set and GAPDH one of the densest; any unqualified statement that all five controls were observed in the same 187 species would be false. The whole-protein conservation test in `results/attempt7c_wholeprotein.json` uses the qualified columns of the retrieved sequences, and its pairwise tests should be read as comparisons within these realized panels. [PENDING: a frozen sensitivity analysis that equalizes species membership across all six proteins before drawing broader evolutionary conclusions.]

12 Fourth-tryptophan model-grid audit

12.1 Pre-registered discriminators and an undefined locked width

The attempt-5b gate in `PREREG_ATTEMPT5B.md` treats the persistence-integral comparison as expected by construction, with no evidential weight by itself. The independent discriminators were peak anisotropy ordering (B2) and magnetic-field discrimination width (B3). On the locked original grid both FWHM fields are encoded as zero in `results/h1_attempt5b_tradeoff.json` because both curves were still rising at the grid edge; these zeros denote *undefined width*, not

a real zero-width biological response. A pre-extension amendment recorded the grid issue and extended the field range before computing the width.

Table 7: Reported quantities from the locked and extended model grids. An undefined original FWHM must not be treated as a pass.

Quantity	tetrad	triad
Original-grid peak proxy	0.001048	0.001049
Original-grid FWHM	undefined	undefined
Extended-grid peak proxy	0.192125	0.192123
Extended-grid FWHM (μT)	217389.2	217389.2

12.2 Interpretation kept at the model’s resolution

The reported persistence-integral ratio is 43.929 under the model’s assigned lifetimes. The extended-grid FWHM values match at the tested interpolation, while the extended-grid peak ordering flips relative to the tiny locked-grid difference; neither observation proves equality over all field strengths or in a living cell. The source JSON calls the original tradeoff direction non-robust and favors a no-difference-detected interpretation. [PENDING: direct measurement of spin dynamics and FWHM in the same constructs before claiming physiological temporal advantage.]

13 Exploratory robustness and negative-control geometry

13.1 Post-hoc perturbations are not the locked gate

The committed `results/h1_attempt5b_perturbation.json` labels its perturbations explicitly exploratory and post-hoc. Its numerical integration reports a reference ratio of 43.971, distinct from the original attempt-5b gate output’s 43.929; these are not interchanged here. Perturbing the rate parameters can show that a large ratio persists under the stipulated model but cannot supply an independent discriminator if the ratio was already expected from the assigned lifetime difference. All rows in the following ledger were computed under that model, not measured in a cell. An opposed 30% perturbation gives a ratio of 23.7 in that file. The persistence

Table 8: Post-hoc lognormal perturbation distribution of the tetrad/triad persistence ratio (committed exploratory file). The threshold of 1.25 belongs to the original gate, but checking it after observing the model does not make this a new locked test.

Lognormal σ	median	fifth percentile	95th percentile	fraction above 1.25
0.05	44.0	39.3	49.3	1.0
0.10	44.1	34.8	55.6	1.0
0.20	44.0	27.5	70.0	1.0
0.30	43.9	21.9	88.3	1.0

result is thus insensitive to these modeled perturbations, while selectivity is still the unresolved, more demanding part of the fourth-tryptophan hypothesis. [PENDING: rate distributions and spin-chemical response measured independently of the model assumptions.]

13.2 Alternative geometric coordinates fail to recover construct ordering

A separate committed exploratory negative control, `results/h1_negative_control_geometry.json`, recomputes the three-construct score with four alternative distance coordinates. Its recorded boolean for recovery of the Xu ordering is false for all four, so choosing a different distance alone does not rescue this particular score. The table lists each coordinate for the European robin, pigeon, and chicken constructs in that order; these are model inputs in angstroms rather than new experimental distances. These alternatives are not independent biological trials. The

Table 9: Model geometric negative-control inputs and test status, directly from the committed JSON.

Coordinate	erCry4	ClCry4	GgCry4	Xu ordering recovered?
terminal	13.76	16.18	16.15	no
terminal-centroid	20.21	21.85	21.81	no
triad	11.06	11.74	11.71	no
third-to-fourth hop	6.09	7.44	7.49	no

numerical score also saturates at several long lifetimes in the negative-control file, further limiting interpretation of ranking ties. [PENDING: an independently calibrated kinetic map between structure and spin-selective lifetime.]

14 Attempt 7d sampling robustness and 7e Cry2 falsification

14.1 What the 7d robustness check tested

The committed `results/attempt7d_robustness.json` reports 10,000 column-bootstrap replicates and 116 leave-one-species-out estimates for Cry1. Its first condition compared the Cry1 interval lower bound with all five original control point estimates; its second required all leave-one-species-out W estimates to remain at least 0.99. Both pass. These are robustness checks for the original contrast, not a Cry1-specific mechanism test. The minimum Cry1 leave-one-

Table 10: Point W and 95% column-bootstrap intervals from the committed 7d file; the Cry2 row is the separate committed 7e control and is not one of the original 7d five controls.

Protein	point W	lower	upper
CRY1	0.99547	0.99151	0.99838
GAPDH	0.98988	0.98515	0.99384
RHO	0.97554	0.96851	0.98148
LDHA	0.97176	0.96168	0.98073
PKM	0.96748	0.95793	0.97612
OPN4	0.89617	0.88224	0.90957
CRY2 (7e)	0.99411	0.98905	0.99789

species-out value is 0.99491 across 116 deletions. This supports stability within the sampled avian panel; it does not remove phylogenetic dependence or show that Cry1's conservation is unique among photoreceptors.

14.2 The photoreceptor-family control defeats specificity

The 7e amendment chose the Cry2 paralog before its outcome. The committed 7e Cry2 control result records 112 recovered of 118 requested species, 203 qualified columns, $W_{\text{Cry2}} = 0.99411$, and the 95% interval [0.98905, 0.99789]. The Cry1 interval lower bound, 0.99151, is below the Cry2 point value; the Cry1 point value, 0.99547, is below Cry2's upper bound. Both locked discrimination clauses are false and win is false. The observed difference between Cry1 and Cry2 does not earn a magnetosensor-specific canalization claim. This falsification supersedes any reading of the five-control 7c win as mechanistic or unique to Cry1.

14.3 Structural transfer boundary

The Cry2 reference is a 400-residue ESMFold crop, with 369 aligned C-alpha pairs and a global superposition RMSD of 6.03 Å. Three tryptophans map; the fourth is absent from this crop. The source notes that this omission can inflate the Cry2 background statistic. It does not license changing the locked 7e outcome after seeing it. [PENDING: prospectively designed full-length Cry2 and additional cryptochrome-family comparators with matched species and coverage.]

15 Attempt-4 frozen gate ledger

15.1 Why a single failed gate is decisive

The physics-forward attempt assessed five pre-specified aspects of the distance-anchored spin model, and the final result file records ALL: false. A pass on the equalized-distance control cannot compensate for failure to recover the experimental interspecies ordering. The following values come from `results/h1_attempt4_gates.json`; each is a computational score or check, not a new independent measurement.

Table 11: All frozen attempt-4 decision fields, including sub-gates; P1 model scores share the model's arbitrary score scale.

Gate	committed decision fields	passes?
P1	robin $S=1.88\text{e-}10$; pigeon $S=3.757\text{e-}09$; chicken $S=3.742\text{e-}09$	no
P2	triad peak=0.001049; tetrad peak=0.001048; triad $S=9.34\text{e-}11$; tetrad $S=1.88\text{e-}10$	no
P3i	mean equalized distance=15.36 angstrom; shrink fraction=1.0	yes
P3ii	anchor order: 0 of 7	no
P4	ordering kept: 0 of 100	no
P5	wide curve reported	yes
ALL	conjunction in result file	no

15.2 Anchor and robustness interpretation

Seven anchor choices were evaluated under P3ii; none recovered the target order. P4's 0/100 sign-consistency finding reinforces that the wrong order is not a single-grid fluctuation under

the implemented perturbations. P5’s “wide curve reported” is a completeness check, not experimental validation. A failed P2 tetrad comparison also prevents promoting a duration result into a general performance win. This ledger describes the specified attempt-4 model and its limits, not all radical-pair models. [PENDING: independently determined interspecies rates and hyperfine parameters for a prospectively locked replication.]

16 Lifetime-to-score grid: the saturating surrogate

16.1 All sampled lifetime points

The committed attempt-4 stage-one result records a shared lifetime-to-anisotropy score curve. Its 16 sampled lifetimes are listed below without interpolation. The model scores rise and saturate across the sampled interval, so increasing a lifetime can increase a score while still ranking the robin below the controls when the distance-to-lifetime mapping itself points the wrong way. This table audits the intermediate transfer function, not a measured spin lifetime.

Table 12: Full sampled $S(\tau)$ grid transcribed from `results/h1_attempt4_stage1.json`. Scores are displayed in 10^{-9} model units. The two side-by-side halves preserve all sixteen computed pairs.

τ (μ s)	$S/10^{-9}$	τ (μ s)	$S/10^{-9}$
0.10	0.09340	1.00	2.27593
0.15	0.18898	1.50	2.89695
0.20	0.29425	2.00	3.23759
0.30	0.56624	3.00	3.57059
0.40	0.85737	4.00	3.72059
0.50	1.14414	5.00	3.79973
0.65	1.54474	7.00	3.87536
0.80	1.89450	10.00	3.91863

16.2 The three-species mapping behind the rank reversal

The same committed stage-one artifact records the terminal distance and modeled tetrad and triad lifetimes for each construct. A larger terminal distance maps to longer modeled tetrad lifetime in this surrogate, placing the two controls near the high-score plateau and the robin near the low end of the curve. These distances are derived from modeled structures; neither a score nor the inferred lifetime is an independent measurement of a living bird.

Three-species stage-one ledger. Distances in angstroms; inferred lifetimes in microseconds.

construct	terminal tetrad	terminal triad	tetrad τ	triad τ
erCry4	13.76	11.06	0.1495	0.0034
ClCry4	16.18	11.74	4.4540	0.0089
GgCry4	16.15	11.71	4.2711	0.0086

The modeled robin tetrad lifetime (0.1495 μ s) is far below the pigeon and chicken estimates (4.454 and 4.2711 μ s). Because the response curve is monotonic on this sampled grid, the surrogate anti-predicts the experimental ordering instead of merely missing a significance threshold. [PENDING: rate constants determined independently of this structural-distance surrogate and tested against additional constructs.]

17 Post-Cry2 falsification ledger and descriptor boundary

17.1 Ecological interaction: neither paralog separates the phenotypes

The 09:03 locked 7f-b gene-by-phenotype arm compared $\Delta W = W_{\text{migratory}} - W_{\text{sedentary}}$ within each gene under 10,000 label permutations. Committed `results/attempt7f_b_gxp.json` reports Cry1 $\Delta W = +0.00142$ ($p = 0.291$) and Cry2 $\Delta W = +0.00038$ ($p = 0.393$). The 112-species matched intersection gives the same negative decision. A difference in numerical signs or magnitudes without the locked threshold cannot be promoted into a gene-by-phenotype interaction. Ecological association is not measured magnetosensitivity; phylogenetic dependence also remains a limit of a label shuffle.

17.2 Radical-pair sphere: conserved in both paralogs

The 7f-d partition included four locked chain sites in 6PTZ numbering (395, 372, 318, 369), plus residues within 5.0 Å of FAD or a chain Trp by heavy atoms. The committed 7f-d result yields 68 qualified Cry1 and 49 Cry2 columns. Cry1's sphere $W = 0.99949$ (95% CI [0.99873, 1.0]) overlaps Cry2's $W = 0.99745$ (95% CI [0.99253, 1.0]). The locked differential-constraint win fails; strong conservation of the sphere is shared, not evidence of magnetosensor specificity.

17.3 Network inversion and C-terminal tail nulls

The 10:13 7g amendment locked a top-200 APC-corrected mutual-information edge budget per gene over qualified variable columns. The statistic was the largest connected component containing a radical-pair-sphere residue. Committed `results/attempt7g_ab.json` has no such component for Cry1 (0; 34 variable columns), versus a 26-residue Cry2 component with five surface residues. The independent 112-species matched-intersection sensitivity in `results/attempt7g_a_sensitivity.json` repeats that inversion: Cry1 0 (41 variable columns), Cry2 26 with five surface sites. This statistic does not identify physical protein interactions or prove the Cry2 network has magnetosensory function.

The 7g-b C-terminal-quarter formal-charge and charged-density tests used 10,000 phenotype-label permutations. Cry1's net-charge and density tests are both $p = 1.0$; Cry2 net charge is $p = 0.0235$ and density $p = 0.32847$. The sole marginal signal is in the wrong paralog for the locked Cry1-specific win. Neither arm rescues specificity; these are static sequence descriptors, not direct electrostatics or measurements of conformational dynamics. [PENDING: independently observed magnetosensitivity against matched full-length paralog sequences, kinetic measurements, and physical interaction tests.]